AI-Based Cost Management System

AI-Powered Invoice Intelligence & Fraud Detection with Federated Learning + LoRA

Huzaifa Ahmed Khan • Muhammad Bilal Adnan • Ali Saeed
Supervisor: Dr. S.M. Faisal Iradat

Institute of Business Administration Karachi
Leadership and Ideas for Tomorrow

Herbion NATURALS

IBA SMCS
School of Mathematics and Computer Science

Problem: Manual Cost Tracking in Enterprise Procurement Workflows

Problem Overview / Background

Core Problem: Organisations lose 5-10% of annual spend to invoice fraud, duplicate payments, and manual entry errors because procurement cost data is locked inside unstructured paper and PDF documents.

Existing Procurement Flow:

Vendor → Paper Invoice → Manual Entry

Day 0 — Day 3-7

Manual keying introduces ~4% error rate; approval cycles take 3–7 days per invoice

Consequence Scenario:

PO Raised → GR Received → Invoice Mismatch

PO-GR-Invoice 3-way match fails silently; duplicate or inflated invoices get paid.

Current Approaches and Limitations

1. ERP Manual Entry (SAP/Oracle): Error-prone, slow, requires trained operators

2. Rule-Based OCR (Tesseract): Fails on scanned/degraded documents; no semantic understanding

3. Commercial AP Automation: High cost; centralises sensitive invoice data on vendor servers

Gap in Existing Solutions

- No open-source tool combines OCR + NLP field extraction + 3-way match in one pipeline
- All existing solutions require raw invoice data centralised - violating financial privacy regulations
- Full fine-tuning of document model needs thousands of labelled invoices, unavailable for most SMEs
- None provide integrated budget forecasting or anomaly detection

Our Goal: A privacy-preserving, compute-efficient AI pipeline using Federated Learning + LoRA that digitises invoices collaboratively without sharing raw data.

Solution: AI-Based Cost Management System

Core System Architecture

Three Main Components:

- **LoRA Fine-Tuning:** Adapts LayoutLM with low-rank weight updates only
- **Federated Learning:** FedAvg aggregates LoRA adapters from N clients
- **Analytics Engine:** 3-way matcher + anomaly detector + Streamlit dashboard

Anomaly Detection - 3 Layers

L1 Rules: duplicate invoice no., amount out of range, suspicious round number >$10k, invoice predates PO, weekend/stale (>90 days)

L2 Isolation Forest: unsupervised ML on amount, unit price & date lag - no fraud labels needed

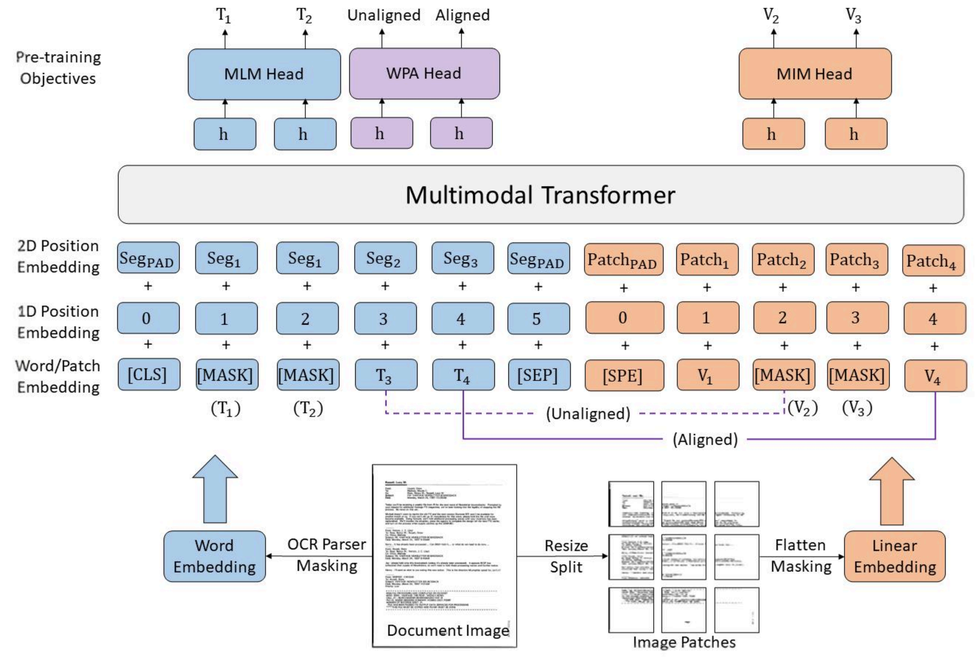
L3 InvoiceAnomalyNet: MLP trained via FL - 7 engineered features, sigmoid >0.5 → FLAGGED

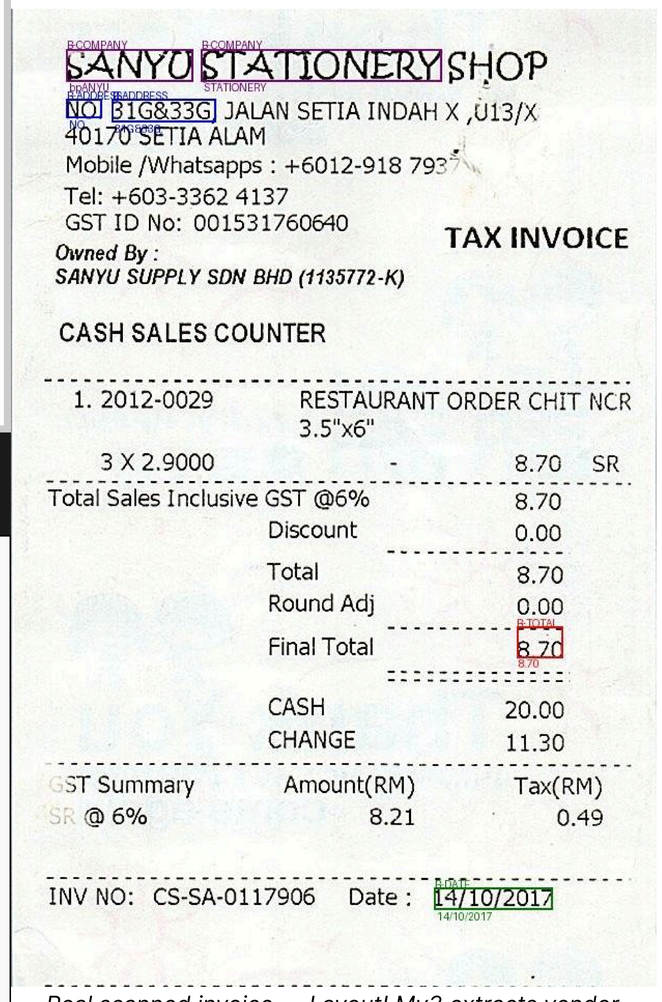
Key Components & Implementation

Four-Tier Extraction Cascade & Pipeline

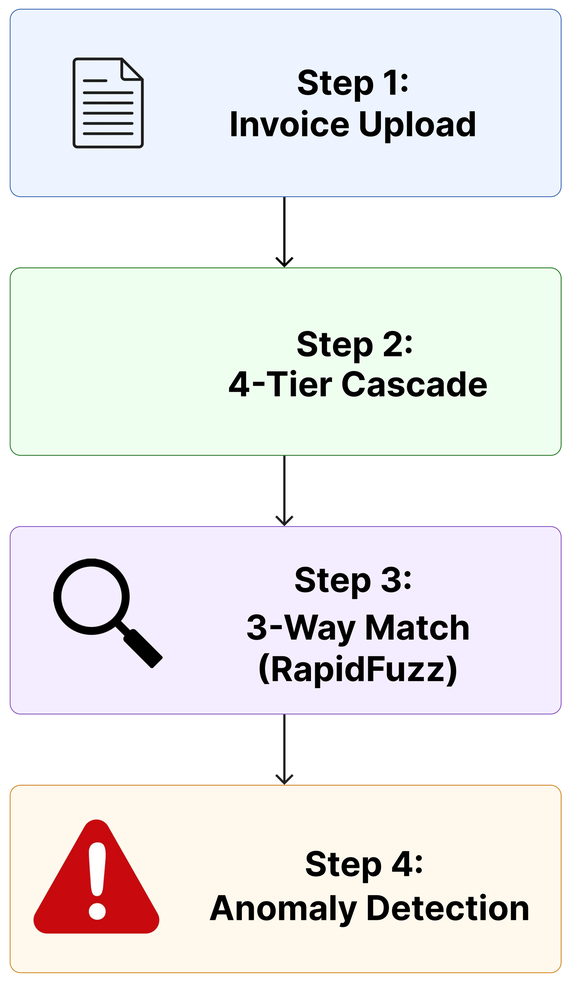
T1	FL-LoRA LayoutLMv3 - federated fine-tuned; highest accuracy on known invoice styles
T2	Base LayoutLMv3 - layout-aware NER without FL adapter; catches unseen layouts
T3	Spatial PDF Parser + Regex - pdfplumber with coordinate-aware field patterns
T4	Tesseract + GOT-OCR 2.0 - universal fallback; always produces output

LayoutLMv3 Architecture & LoRA Adaptation





Real scanned invoice — LayoutLMv3 extracts vendor, date, totals & invoice no.



AI-Powered Invoice Processing & Anomaly Detection Pipeline

Results & Evaluation

Extraction & Matching:

- FL+LoRA final accuracy: 87.9% (avg. F1 across 6 fields)
- Centralised fine-tune: 92.9% - privacy gap only 5.4%
- Best field INVOICE_NO: 0.966 (FL) vs 0.999 (central)

FL & LoRA Efficiency:

- Trainable params: 294K of 110M (LoRA r=8)
- Comm: 1,152 KB/round vs 1.2 GB full FT (1,066x reduction)
- Raw invoice data never leaves client

Field	FL+LoRA	Centralised FT
DATE	0.805	0.849
INVOICE_NO	0.966	0.999
PO_NO	0.951	1.000
SUBTOTAL	0.927	0.970
TOTAL	0.836	0.896
VENDOR	0.921	0.986
Final Accuracy	**0.879**	**0.929**

Conclusion & Future Work

Key Takeaway:

AI-Cost Management System combines a 4-tier extraction cascade, FL+LoRA fine-tuning (294K params, 1,066x less communication), and a 3-layer anomaly detector (Rules + Isolation Forest + InvoiceAnomalyNet) to achieve 87.9% field F1, only 5.4% below centralised while keeping all raw invoice data on-device.

Future Work:

- Tighten DP: current e~8.74/round → target e<5 via subsampling & sigma tuning
- Multilingual: French+English → Arabic, Urdu, Asian scripts
- Real-time streaming: process invoices from email/ERP

[1] Xu et al. (2020). LayoutLM: Pre-training of Text and Layout for Document Image Understanding. KDD 2020, pp. 1192-1200.

[2] Hu et al. (2022). LoRA: Low-Rank Adaptation of Large Language Models. ICLR 2022. arXiv:2106.09685.

[3] Zhang et al. (2020). FedOCR: Communication-Efficient Federated Learning for Scene Text Recognition. arXiv:2007.11462.